

Continuum Mechanics
class # 02
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Section 1.3 Second-Order Tensors

Defn: By a second order tensor T on the vector space $\mathcal{V}$ we mean a mapping $T: \mathcal{V} \rightarrow \mathcal{V}$ which is linear:

$$1) T(\vec{u} + \vec{v}) = T\vec{u} + T\vec{v}, \quad \forall \vec{u}, \vec{v} \in \mathcal{V}$$

$$2) T(\alpha \vec{u}) = \alpha T\vec{u}, \quad \forall \alpha \in \mathbb{R}, \vec{u} \in \mathcal{V}$$

The zero tensor, $\mathbf{0}$, is the tensor with the property that $\mathbf{0}\vec{v} = \vec{0} \in \mathcal{V}$ for each $\vec{v} \in \mathcal{V}$.

The identity tensor, $\mathbf{I}$, is the tensor with the property that $\mathbf{I}\vec{v} = \vec{v}$ for all $\vec{v} \in \mathcal{V}$.

Two second-order tensors, T and S , say, are said to be equal iff

$$T\vec{v} = S\vec{v}, \quad \forall \vec{v} \in \mathcal{V}. \quad //$$

Example: (Projection) Suppose that $\vec{e} \in \mathcal{V}$ is a unit vector, i.e., $\vec{e} \cdot \vec{e} = 1$.

Define, for all $\vec{v} \in \mathcal{V}$

$$P\vec{v} = (\vec{v} \cdot \vec{e}) \vec{e}$$

P is called the projection onto $\vec{e}$. Then P is a second order tensor.

Example (Rotation): Suppose we wish to rotate any vector $\vec{v} \in V$ through a given angle θ about a fixed axis $\vec{e}_3$, where

$$\mathcal{F} = \{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$$

where $\mathcal{F}$ is a given frame. Define

$$S\vec{e}_1 = \cos(\theta)\vec{e}_1 + \sin(\theta)\vec{e}_2$$

$$S\vec{e}_2 = \cos(\theta)\vec{e}_2 - \sin(\theta)\vec{e}_1$$

$$S\vec{e}_3 = \vec{e}_3$$

Why does this completely determine the second-order tensor S ? //

Example: (Reflection): Suppose that M is the unique plane passing through the origin with the unit normal $\vec{e} \in V$. Here

$$\vec{e} \cdot \vec{e} = 1$$

and

$$\vec{v} \cdot \vec{e} = 0, \quad \forall \vec{v} \in M.$$

Set

$$T\vec{v} = \vec{v} - 2(\vec{v} \cdot \vec{e})\vec{e},$$

T is a second order tensor.

What is

$$T(T(\vec{v})) \stackrel{?}{=}$$

//

Exercise: Create a second order tensor that projects the image of $\vec{v}$ onto the same plane M . What are the properties of this tensor? //

Tensor Algebra

Definition: The set of all second-order tensors is denoted V^2 . Suppose $S, T \in V^2$ and $\alpha \in \mathbb{R}$. Then $S+T$ is the object defined by the mapping

$$(S+T)(\vec{v}) = S\vec{v} + T\vec{v}, \quad \forall \vec{v} \in V.$$

The object αT is defined by the mapping

$$(\alpha T)(\vec{v}) = \alpha(T\vec{v}), \quad \forall \vec{v} \in V.$$

It is not hard to show that

$$S+T \in V^2 \quad \alpha T \in V^2$$

Definition: Suppose that $S, T \in V^2$. The product of S and T is the object ST defined by

$$(ST)(\vec{v}) = S(T\vec{v}), \quad \forall v \in V.$$

One can show that $ST \in V^2$!

This shows that V^2 is a "vector space" or a real linear space.

Representation in Given Frames

Definition Suppose that $\mathcal{F} = \{\vec{e}_i\}$ is a coordinate frame and $S \in V^2$. The coordinates of S are the nine scalars

$$S_{i,j} = \vec{e}_i \cdot S \vec{e}_j, \quad 1 \leq i, j \leq 3. //$$

Proposition: Suppose the $\mathcal{F} = \{\vec{e}_i\}$ is a frame, $\vec{v} = v_i \vec{e}_i \in V$, and $S \in V^2$. Set

$$\vec{u} = S \vec{v}.$$

Then

$$(1.6) \quad u_i = S_{i,j} v_j, \quad (\text{sum over } j)$$

where u_i , $1 \leq i \leq 3$, are the coordinates of $\vec{u}$ in $\mathcal{F}$.

Proof: By definition

$$u_m \vec{e}_m = \vec{u} = S(\vec{v}) = S(v_n \vec{e}_n).$$

Since S is linear

$$\begin{aligned} u_m \vec{e}_m &= S(v_n \vec{e}_n) \\ &= v_n S \vec{e}_n \end{aligned}$$

Take the dot product with $\vec{e}_k$.

$$\vec{e}_k \cdot (u_m \vec{e}_m) = \vec{e}_k \cdot (v_n S \vec{e}_n)$$

Observe that, since the dot product is linear

$$\begin{aligned} \vec{e}_k \cdot (u_m \vec{e}_m) &= u_m (\vec{e}_k \cdot \vec{e}_m) \\ &= u_m \delta_{k,m} \\ &= u_k. \end{aligned}$$

Similarly,

$$\begin{aligned} \vec{e}_k \cdot (v_n S \vec{e}_n) &= v_n \vec{e}_k \cdot (S \vec{e}_n) \\ &= v_n S_{kn} \\ &= S_{kn} v_n \end{aligned}$$

Thus,

$$u_k = S_{kn} v_n \quad //$$

Notation: When I want to extract i, j components from $S \in \mathbb{R}^2$, with respect to a frame $\mathcal{F} = \{\vec{e}_i\}$, I write

$$[S]_{i,j} = S_{i,j} = \vec{e}_i \cdot (S \vec{e}_j)$$

We will also commonly write S as a matrix

$$[S] = \begin{pmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

We call $[S]$ the matrix representation of $S \in V^2$ in the frame $\mathcal{F} = \{\vec{e}_i\}$.

In matrix form $[S]_{i,j}$ refers to component in the i^{th} row and j^{th} column, as usual.

Definition: let $S \in V^2$ and $[S]$ is matrix representation in the frame $\mathcal{F}$. The transpose of $[S]$, denoted $[S]^T$ is the matrix in $\mathbb{R}^{3 \times 3}$

$$[S]^T = \begin{pmatrix} S_{11} & S_{21} & S_{31} \\ S_{12} & S_{22} & S_{32} \\ S_{13} & S_{23} & S_{33} \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

In other words,

$$[S]^T_{i,j} = [S]_{i,j}.$$

Exercise: let $S \in V^2$ and suppose $\mathcal{F} = \{\vec{e}_i\}$ is a given frame. Then

$$[S] = \begin{pmatrix} | & | & | \\ [S\vec{e}_1] & [S\vec{e}_2] & [S\vec{e}_3] \\ | & | & | \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

let $\vec{u}, \vec{v} \in V$ satisfy

$$\vec{u} = S\vec{v}.$$

Then

$$[\vec{u}] = [S][\vec{v}]$$

↑ matrix-vector multiplication
from linear algebra.

Recall that $[\vec{u}]$ is the coordinate column vector with respect to $\mathcal{H} = \{\vec{e}_i\}$. //

Example: let S be the rotation tensor introduced earlier. Then

$$[S] = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Example: Consider the identity tensor $I\vec{v} = \vec{v}$, for all $\vec{v} \in V$. Let $\mathcal{H} = \{\vec{e}_i\}$ be any frame. Then

$$I_{i,j} = \delta_{i,j}$$

To see that this is so,

$$\begin{aligned} I_{i,j} &= \vec{e}_i \cdot (I \vec{e}_j) \\ &= \vec{e}_i \cdot \vec{e}_j \\ &= \delta_{i,j} \end{aligned}$$

Dyadic Products:

Defn: let $\vec{a}, \vec{b} \in V$ be given the second-order tensor $\vec{a} \otimes \vec{b}$, defined by

$$(\vec{a} \otimes \vec{b}) \vec{v} = (\vec{b} \cdot \vec{v}) \vec{a}, \quad \forall \vec{v} \in V$$

is called the dyadic product of $\vec{a}$ and $\vec{b}$.

Proposition: Suppose that $\mathcal{H} = \{\vec{e}_i\}$ is a frame and $\vec{a} = a_i \vec{e}_i$ $\vec{b} = b_j \vec{e}_j$. Then

$$[\vec{a} \otimes \vec{b}]_{ij} = a_i b_j$$

Proof: By definition

$$\begin{aligned} [\vec{a} \otimes \vec{b}]_{ij} &= \vec{e}_i \cdot (\vec{a} \otimes \vec{b}) \vec{e}_j \\ &= \vec{e}_i \cdot (\vec{b} \cdot \vec{e}_j) \vec{a} \\ &= \vec{e}_i \cdot (b_j \vec{a}) \\ &= b_j (\vec{e}_i \cdot \vec{a}) \\ &= b_j a_i \\ &= a_i b_j. \quad // \end{aligned}$$

Exercise: Show that for a given frame $\mathcal{H} = \{\vec{e}_i\}$

$$[\vec{a} \otimes \vec{b}] = [\vec{a}] [\vec{b}]^T$$

Here the transpose turns the column vector into a row vector.

$$\begin{matrix} [\vec{a}] & [\vec{b}]^T & \in \mathbb{R}^{3 \times 3} \\ 3 \times 1 & 1 \times 3 & \end{matrix}$$

Proposition: let $S \in V^2$ be given and suppose $\mathcal{F} = \{\vec{e}_i\}$ is any frame. Define, as usual

$$S_{ij} = [S]_{i,j} = \vec{e}_i \cdot (S\vec{e}_j)$$

Then

$$(1.7) \quad S = S_{ij}(\vec{e}_i \otimes \vec{e}_j). \quad (2 \text{ sums})$$

Proof: let $\vec{u} \in V$ be arbitrary. Set

$$\vec{v} := S\vec{u}$$

Assume that $\vec{v} = v_m \vec{e}_m$, $\vec{u} = u_r \vec{e}_r$.

$$\begin{aligned} S_{ij}(\vec{e}_i \otimes \vec{e}_j)\vec{u} &= S_{ij}(\vec{e}_i \otimes \vec{e}_j)u_r \vec{e}_r \\ &= S_{ij}u_r((\vec{e}_i \otimes \vec{e}_j)\vec{e}_r) \\ &= S_{ij}u_r(\langle \vec{e}_j, \vec{e}_r \rangle \vec{e}_i) \\ &= S_{ij}u_r(\delta_{jr} \vec{e}_i) \\ &= S_{ij}u_j \vec{e}_i \\ &= v_i \vec{e}_i \\ &= S\vec{u}. \end{aligned}$$

Thus

$$S = S_{ij}(\vec{e}_i \otimes \vec{e}_j). \quad \square$$

Second-Order Tensor Algebra in Components:

It is not hard to show that, if $S, T \in V^2$ and $\mathcal{F} = \{\vec{e}_i\}$ is a frame,

$$[S+T] = [S] + [T]$$

and

$$[\alpha T] = \alpha [T]$$

where $[\cdot]$ means the matrix representation with respect to the frame.

Exercise: let $\mathcal{F} = \{\vec{e}_i\}$ be a frame. Suppose that $S, T \in V^2$. Set $U = ST$. Then

$$U_{i,j} = S_{i,k} T_{k,j}, \quad 1 \leq i, j \leq 3.$$

To prove this result, let us first show the following

Proposition: For any $\vec{a}, \vec{b}, \vec{c}, \vec{d} \in V$

$$(\vec{a} \otimes \vec{b})(\vec{c} \otimes \vec{d}) = (\vec{b} \cdot \vec{c}) \vec{a} \otimes \vec{d}$$

Proof: let $\vec{v} \in V$ be arbitrary. Then

$$\begin{aligned} ((\vec{a} \otimes \vec{b})(\vec{c} \otimes \vec{d}))\vec{v} &= (\vec{a} \otimes \vec{b})(\vec{c} \otimes \vec{d})\vec{v} \\ &= (\vec{a} \otimes \vec{b})(\vec{d} \cdot \vec{v})\vec{c} \\ &= (\vec{d} \cdot \vec{v})(\vec{a} \otimes \vec{b})\vec{c} \\ &= (\vec{d} \cdot \vec{v})(\vec{b} \cdot \vec{c})\vec{a} \end{aligned}$$

$$\begin{aligned}
&= (\vec{b} \cdot \vec{c}) (\vec{a} \cdot \vec{v}) \vec{a} \\
&= (\vec{b} \cdot \vec{c}) ((\vec{a} \otimes \vec{a}) \vec{v}) \\
&= ((\vec{b} \cdot \vec{c}) (\vec{a} \otimes \vec{a})) \vec{v}.
\end{aligned}$$

Since $\vec{v}$ is arbitrary, the result is proven. //

Now, we can finish off the previous exercise.

Using (1.7) and the last proposition,

$$\begin{aligned}
U_{i,l} \vec{e}_i \otimes \vec{e}_l &= U \\
&= (S_{ij} \vec{e}_i \otimes \vec{e}_j) (T_{kl} \vec{e}_k \otimes \vec{e}_l) \\
&= S_{ij} T_{kl} ((\vec{e}_i \otimes \vec{e}_j) (\vec{e}_k \otimes \vec{e}_l)) \\
&= S_{ij} T_{kl} ((\vec{e}_j \cdot \vec{e}_k) (\vec{e}_i \otimes \vec{e}_l)) \\
&= S_{ij} T_{kl} (S_{jk} (\vec{e}_i \otimes \vec{e}_l)) \\
&= S_{ik} T_{kl} (\vec{e}_i \otimes \vec{e}_l)
\end{aligned}$$

Thus

$$U_{i,l} = S_{ik} T_{kl} //$$